

CSE 106

Online on Graphs (A1/A2)

There are **N** courses numbered from **1** to **N**. Some courses have prerequisite requirements, meaning you must complete certain courses before enrolling in them. These prerequisites are represented as directed edges, where an edge from course **A** to course **B** indicates that course **A** must be completed before course **B**.

You will receive multiple queries. Each query provides a specific course, and your task is to determine the number of prerequisite courses that must be completed before enrolling in that course.

Input

- The first line contains two integers **N** and **M**, the number of courses and the number of prerequisite relationships.
- The next **M** lines each contain two integers **A** and **B**, meaning that course **A** is a prerequisite for course **B**.
- The following line contains an integer **Q**, the number of queries.
- The next **Q** lines each contain a single integer **C**, representing a course for which you need to determine the prerequisite count.

Output

For each query, print a single integer, representing the number of prerequisite courses required for the given course.

Input	Output
5 5 1 2 1 3 3 4 4 5 2 5 2 5 4	4 2